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Title: Bias and Ethics of AI Systems Applied in Auditing - A Systematic Review

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W. Murikah, J. K. Nthenge, and F. M. Musyoka, "Bias and Ethics of AI Systems Applied in Auditing - A Systematic Review," in *Scientific African*, vol. 25, art. no. e02281, Sep. 2024, doi: 10.1016/j.sciaf.2024.e02281.

Summary:

This systematic review examines the integration of artificial intelligence (AI) into auditing practices, highlighting its potential to automate processes and extract insights from complex datasets while underscoring the profound ethical challenges it introduces. The authors analyze biases in AI systems, including algorithmic prejudices, lack of transparency, accountability issues, and fairness concerns, which can undermine audit reliability and lead to discriminatory outcomes. Through a rigorous methodology, the study reviews 83 high-quality publications from 2018 to 2023 across disciplines like computer science, accounting, and ethics, identifying key sources of bias such as data deficiencies (e.g., incomplete or skewed datasets), demographic homogeneity in training data, spurious correlations that mislead models, improper comparators in algorithmic decisions, and human cognitive biases amplified by AI. These factors create downstream risks, including trade-offs between operational efficiency and thorough due diligence, erosion of auditors' professional judgment and skills due to over-reliance on automation, heightened data dependency vulnerabilities, and privacy breaches from unchecked exploitation of personal information.

The paper also delves into broader ethical implications, such as how AI could exacerbate inequalities in auditing outcomes or compromise stakeholder trust if not managed properly. It discusses real-world interactions where biased AI might perpetuate systemic issues, like favoring certain demographic groups in financial assessments or failing to detect fraud in underrepresented scenarios. To mitigate these, the authors propose a range of technical and ethical safeguards, including causal modeling techniques to reveal hidden biases, comprehensive testing of algorithms with diverse datasets to ensure fairness, regular independent audits of AI systems, and mandatory human oversight to complement automated processes. Additionally, they advocate embedding core ethical principles such as accountability, transparency, and inclusivity directly into AI design from the outset, drawing from interdisciplinary best practices.

In conclusion, the review positions auditors as pivotal guardians in the ethical deployment of AI, emphasizing their role in evaluating systems for societal benefit. Recommendations include establishing robust governance frameworks, conducting pre-deployment risk assessments, implementing continuous monitoring protocols, and developing policies that promote collaboration between technologists, ethicists, and auditors. By translating abstract ethical principles into concrete auditing practices, the study argues, the profession can harness AI's advantages while minimizing harms, fostering greater trust and equity in financial oversight. This work serves as a call to action for stakeholders to prioritize responsible AI innovation in auditing.

